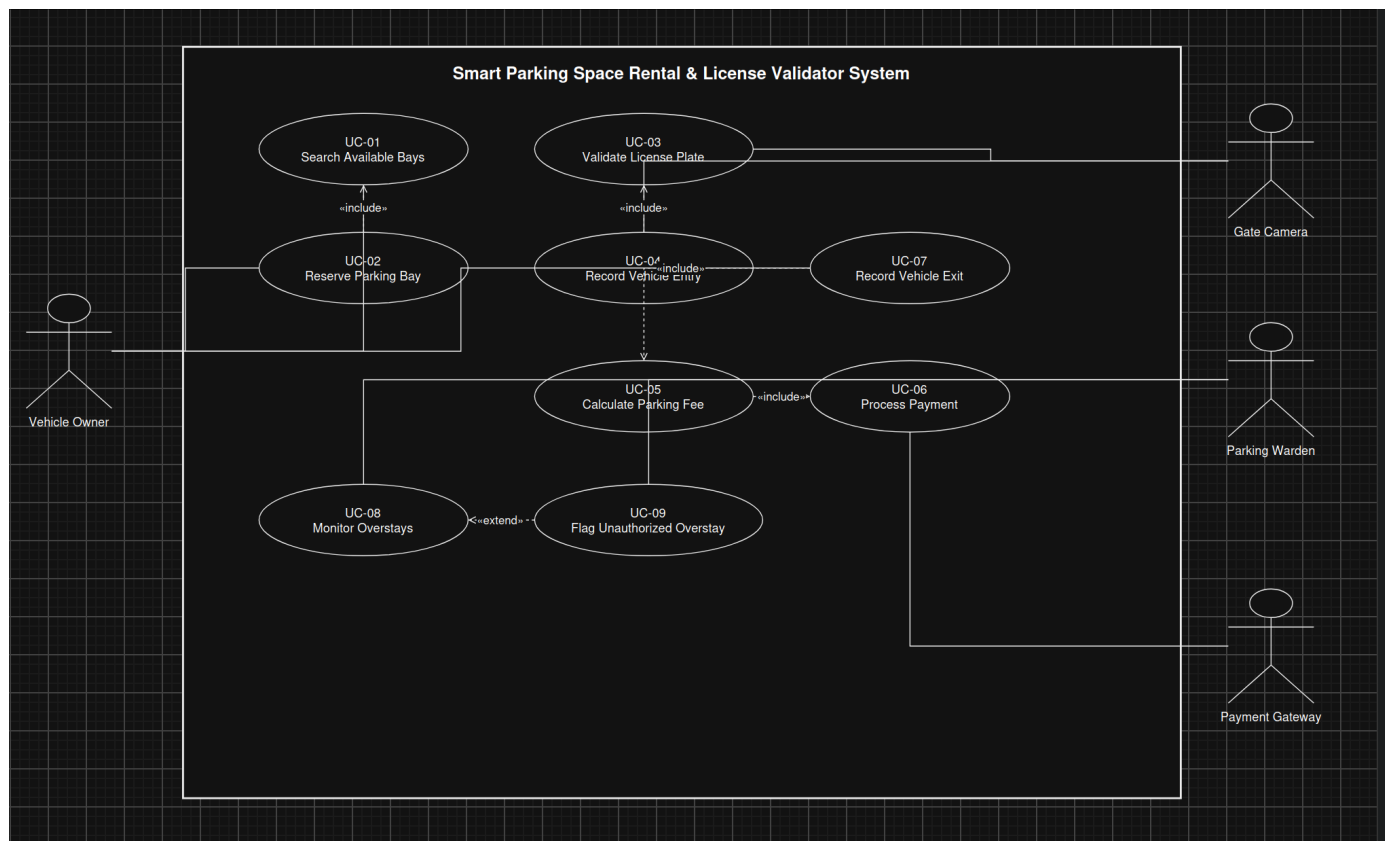


Smart Parking Space Rental & License Validator

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record vehicle entry time and vehicle number plate through a gate camera license plate scan.	High	Pass: For at least 99% of valid test entries, the recorded number plate matches the test vehicle and the entry timestamp is stored automatically. Fail: A valid entry is recorded with an incorrect plate or missing timestamp.	Provides an accurate basis for reservation validation, duration tracking, and billing.
FR-002	Functional	The system shall allow a vehicle owner to search available parking bays and reserve an available bay for a selected time period.	High	Pass: When an available bay and valid time period are selected, the system creates a unique reservation and marks the bay as reserved for that period. Fail: The system creates two active reservations for the same bay and overlapping time period.	Enables real-time parking reservations and prevents double booking.
FR-003	Functional	The system shall calculate the parking fee at vehicle exit using the configured progressive hourly rate tiers and recorded parking duration.	High	Pass: For predefined test durations, the calculated fee exactly matches the configured tiered-rate schedule. Fail: Any billing tier is applied incorrectly or the duration is calculated from an incorrect entry time.	Ensures accurate and transparent billing based on parking duration.
FR-004	Functional	The system shall validate a vehicle's license plate against its active parking reservation when the vehicle enters the parking facility.	High	Pass: A vehicle with a matching active reservation is granted valid-entry status, while a vehicle without a matching active reservation is rejected or flagged for authorization review.	Prevents unauthorized vehicles from using reserved parking spaces.
FR-005	Functional	The system shall allow the Parking Warden to view and flag vehicles that exceed their authorized parking duration.	High	Pass: When a vehicle exceeds its reservation or permitted stay by the configured threshold, it appears in the warden's overstay list with its plate number, bay, and recorded times.	Helps parking staff identify and manage unauthorized overstays.
NFR-001	Nonfunctional — Performance & Security	The gate camera license plate lookup API shall verify parking reservation validity in under 300 ms.	High	Pass: Under simulated peak load, at least 95% of valid reservation-lookup requests complete within 300 ms and meet the configured security controls. Fail: More than 5% exceed 300 ms.	Supports rapid gate operation while maintaining secure reservation validation.
NFR-002	Nonfunctional — Availability & Reliability	The parking reservation and bay-status service shall maintain at least 99.5% monthly availability during operating hours.	High	Pass: Monitoring records at least 99.5% availability during each calendar month, excluding documented scheduled maintenance. Fail: Availability falls below 99.5%.	Ensures customers and parking staff can reliably access real-time bay availability and reservations.



Use-Case Flow Specification — Reserve Parking Bay

Use Case ID	UC-02
Use Case Name	Reserve Parking Bay
Primary Actor	Vehicle Owner
Supporting Actor	Parking System / Gate Camera
Preconditions	The vehicle owner has access to the parking system; parking bay availability data is current; the owner can provide a valid vehicle number plate and required reservation details.
Postconditions	A unique reservation is created for an available bay and time period, the reservation is stored against the vehicle number plate, and the selected bay is unavailable to other reservations for the reserved period.

Main Success Scenario

1. The Vehicle Owner enters the desired parking location, date, and expected parking period.
2. The system retrieves and displays parking bays available for the requested period.
3. The Vehicle Owner selects an available bay and provides the vehicle number plate.
4. The system validates that the selected bay is still available and that the vehicle number plate is valid.
5. The system displays the applicable parking rate and estimated fee for the selected duration.
6. The Vehicle Owner confirms the reservation.
7. The system creates a unique reservation ID and records the bay, vehicle number plate, reservation period, and rate information.
8. The system marks the selected bay as reserved for the requested period and displays the reservation confirmation.

Alternate Flow — Selected Bay Becomes Unavailable

9. At Step 4, the system determines that the selected bay has been reserved by another vehicle or is no longer available.
10. The system does not create the reservation.
11. The system informs the Vehicle Owner that the selected bay is unavailable and refreshes the available-bay list.
12. The Vehicle Owner selects another available bay and returns to Step 5, or cancels the reservation process.

Key Acceptance Checks

- A confirmed reservation receives a unique reservation ID.
- The system does not allow overlapping active reservations for the same bay.
- The reservation stores the vehicle number plate, bay, and reservation period.

- A bay that becomes unavailable before confirmation cannot be reserved.